

Jordan Low Jun Yi

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EDUCATION

National University of Singapore, Grade: 5.0/5.0

Aug 2025 - Aug 2027

Engineering Scholars Programme

Bachelor of Engineering (Biomedical Engineering)

Second Major in Computing (Design and Engineering)

Minor in Artificial Intelligence (Design and Engineering)

Minor in Innovation and Design (iDP)

Specialisation in Tissue Engineering

Specialisation in Robotics

UTown College Programme – Residential College 4 (RC4)

Hanyang University, South Korea, Winter Exchange Programme

Dec 2025 - Jan 2026

Engineering Ethics, and Korean Language

National University of Singapore, Non-Graduating Programme

Aug 2021 - Aug 2025

Life Sciences, Physics, and Computing

NUS High School of Mathematics and Science, Grade: 4.9/5.0

Jan 2017 - Dec 2022

Only student in the cohort to earn Triple Distinction with Honours in Chemistry, Biology and Physics

WORK EXPERIENCE

Agency for Science, Technology and Research (A*STAR)

May 2026 - Present

Innovation and Enterprise Intern (Innovis) | Institute of Molecular and Cell Biology Intern (Proteos)

- Hold concurrent appointments across A*STAR's technology translation arm and its flagship cell and molecular biology institute, bridging laboratory research and commercial deployment.

NUS Yong Loo Lin School of Medicine

Jun 2024 - Present

Research Staff

- Built 3D liver organoid models of metabolic dysfunction-associated steatohepatitis and showed that sirolimus cleared steatotic features, restored actin cytoskeleton organisation and mitochondrial count in diseased hepatocytes, and reduced intracellular lipid volume.
- Ran a genome-wide association study of the triglyceride-to-HDL ratio across 424,032 UK Biobank participants, mapping 221 genomic risk loci with FUMA and MAGMA (LPL, CETP, APOA5), identifying hepatocellular carcinoma pathway enrichment, and linking variants to incident myocardial infarction using Cox proportional-hazards models.
- Trained an EfficientNet-B6 model to stage liver disease from B-mode ultrasound, predicting controlled attenuation parameter and liver stiffness and incorporating FAST and Agile-3 scores. Trained across four Japanese centres and externally validated on a Chinese cohort, reaching AUCs of 0.98 for ruling out MASH and 0.91 for ruling out advanced fibrosis.
- Co-authored four peer-reviewed papers in Heart, Metabolism, Atherosclerosis and npj Gut and Liver.

MDS-Plus

Nov 2024 - Feb 2025

Mechanical Designer Intern

- Ran thermal simulations in SOLIDWORKS for Rover-X, the HomeTeam robotic surveillance dog, improving heat dissipation through iterative redesign of the enclosure.
- 3D-printed customised endoscope prototypes for deploying a colonic support stent and ran loading simulations to assess stent strength under physiological conditions.
- Led project ideation and magnetic simulation in COMSOL Multiphysics for Endopill, a magnetically triggered drug-delivery pill.

BIOMEDICAL ENGINEERING PROJECTS

FreeStride: Lymphedema Therapy Device

2025 - Present

Founder, NUS VIP Grant (S\$10,000)

- Won Champion at IDEATE 2025 with a portable alternative to intermittent pneumatic compression, then pivoted the device to replicate manual lymphatic drainage, a therapy costing up to S\$200 per session and rarely covered by insurance.

KneedIt: Smart Knee Sleeve for Osteoarthritis

2025 - Present

Founder, NUS Medtech Project Director

- Designed a sensor sleeve tracking knee range of motion, EMG and PPG with on-sleeve LED feedback, paired with a companion app that locks the phone past a screen-time threshold until the sleeve verifies rehabilitation exercise.
- Targets the roughly 40% of Singaporeans over 70 living with osteoarthritis, with a caregiver dashboard for adherence monitoring.
- Coded a live prototype at kneed-it2.vercel.app.

COMPUTATIONAL AND MACHINE LEARNING PROJECTS

CARE-Liver: Fibrosis Staging from MRI

2026 - Present

MICCAI CARE-Liver Challenge, Digital Heart Lab

- Built a two-stage pipeline that fine-tuned TotalSegmentator-MRI in nnU-Net from only 30 scans, reaching a Dice score of 0.97.
- Ensembled per-sequence 3D ResNets with missing-modality-robust late fusion to stage fibrosis under incomplete imaging protocols.

Genome-Wide Association Study of the Triglyceride-to-HDL Ratio

2025 - Present

NUS Yong Loo Lin School of Medicine

- Analysed 424,032 UK Biobank participants and mapped 221 genomic risk loci using FUMA and MAGMA, including LPL, CETP and APOA5.
- Identified hepatocellular carcinoma pathway enrichment and linked variants to incident myocardial infarction using Cox proportional-hazards models.

PUBLICATIONS

- Prognosis of patients with ST-elevation myocardial infarction presenting to the emergency department without chest pain: a retrospective nationwide cohort study. *Heart*, 2026. doi.org/10.1136/heartjnl-2025-327601 (PMID 42315345)
- Diversity in metabolic dysfunction-associated steatotic liver disease: a framework for precision medicine. *npj Gut and Liver*, 2026. doi.org/10.1038/s44355-026-00061-3
- Fibro-inflammatory liver injury, not steatosis, predicts incident composite neuropsychiatric events and mortality in the UK Biobank. *Metabolism*, 2026. doi.org/10.1016/j.metabol.2026.156635 (PMID 42150669)
- Organoid models reveal mechanistic connections and sirolimus efficacy in liver-vascular steatosis and foam cell formation. *Atherosclerosis*, 2026. doi.org/10.1016/j.atherosclerosis.2026.120762

UI/UX AND WEB DESIGN

- Kosmode Health (kosmodehealth.com), the public site for an NUS deep-tech nutrition and biomedical startup.
- NUS DE-Scholars (nusdescholars.com), the programme site for the Design and Engineering scholars community.
- KneedIt (kneed-it2.vercel.app), the live product prototype and caregiver dashboard.
- Personal portfolio (jordanlowjunyi.github.io/Jordan).

LEADERSHIP & CO-CURRICULAR ACTIVITIES

NUS RC4 | EXCO for Orcabakes and RC4 Tea Interest Groups

Feb 2026 - Present

Vice Head for RC\$ Tea and Treasurer for Orcabakes

- Directed resource allocation and operational planning across both the interest groups' initiatives, optimising scheduling and capacity under competing constraints.
- Coordinated cross-functional logistics and administrative workflows, ensuring execution fidelity across multiple concurrent workstreams.

NUS RC4 | House of Aquila Programmes Committee

Aug 2025 - May 2026

- Planned and executed house-level events including bonding activities and festive celebrations.
- Co-led planning and execution of Prank Week, a flagship house tradition and major community event requiring coordination across four organisers, tight scheduling, and dynamic problem-solving as conditions changed.
- Oversaw logistics across parallel workstreams, developing adaptive contingency plans to maintain execution quality under uncertainty.
- Successfully fostered participation and enthusiasm across the house, strengthening community culture and event legacy.

NUS Biomedical Engineering | Orientation Group Leader

Aug 2025

- Led an orientation group for incoming Biomedical Engineering students, running activities and supporting their transition into university.

VOLUNTEERING

- NUS Todddycats, nature conservation, Jan 2024 to Aug 2025.
- Fei Yue Community Services, Aug 2024 to Oct 2025.
- Active Aging Advocates, Jan 2023 to Dec 2023.
- Ling Kwang Home and All Saints Home, volunteer staff, Mar 2022 to Sep 2022.
- Lions Befrienders, Team Lead, Jul 2021 to Nov 2021.
- SHINE Children and Youth Services, student volunteer, Jan 2021 to Jul 2021.

AWARDS

- Innovation and Research Award, High Achievement (2026)
- IDEATE, Champion (Oct 2025)
- WorldQuant BRAIN Alphathon, 6th Place (September 2024)
- Tan Kah Kee Young Inventors' Award, Silver (Jul 2022)
- Global Youth Science and Technology Bowl, Grand Prize (Gold) (May 2022)
- International Elementz Fair, Gold (Apr 2022)
- Singapore Science and Engineering Fair, Gold (Mar 2022)
- Singapore Science and Engineering Fair, Silver (Mar 2022)
- A*STAR Young Researcher Award, A*STAR Singapore (Jan 2022)
- National Robotics Competition, 2nd Runner Up (Sep 2021)
- International Chemistry Tournament, Champion (Aug 2021)
- Singapore Chemistry Olympiad, Gold (Jul 2021)
- NUS Chemical Engineering Challenge, 2nd Runner Up (2021)
- A*STAR Science Award (Junior College) (Jan 2021)
- UK Intermediate Biology Olympiad, Gold (Mar 2020)
- Singapore Junior Biology Olympiad, Gold (May 2019)
- Singapore Junior Chemistry Olympiad, Gold (May 2019)
- UK Biology Challenge, Gold (Mar 2019)
- Lifesaving & Bronze Medallion Programme (Mar 2019)